THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

May 11, 2018 TIME: 9:30a.m.-12:00noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL TEN questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.
- Some formulae you may need:

For $n \geq 2$:

$$\int \cos^n \theta \, d\theta = \frac{1}{n} \cos^{n-1} \theta \sin \theta + \frac{n-1}{n} \int \cos^{n-2} \theta \, d\theta$$

$$\int \sin^n \theta \, d\theta = -\frac{1}{n} \sin^{n-1} \theta \cos \theta + \frac{n-1}{n} \int \sin^{n-2} \theta \, d\theta$$

1. (10 marks) Give answers only to the following questions. The process of calculating is **NOT** required.

   (a) Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & 1 & \beta & 0 \end{array}\right).$$

   i. Is it possible for the system to be inconsistent? Explain.

   ii. For what values of β will the system have infinitely many solutions?

   (b) Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right).$$

   i. For what values of a and b will the system have infinitely many solutions?

   ii. For what values of a and b will the system be inconsistent?

   iii. For what values of a and b will the system have unique solution?

2. (10 marks) Determine whether the function

$$f(x,y) = \begin{cases} \dfrac{\sqrt{xy+1}-1}{xy} & x \neq 0 \text{ and y} \neq 0 \text{ and xy} \geq -1, \\ 0 & (x,y) = (0,0). \end{cases}$$

is continuous at $(0,0)$ or not. Prove your result.

3. (10 marks) Find the absolute minimum and absolute maximum value of the function for $f(x,y) = x^2y(4-x-y)$ in the closed region bounded by $x-$axis, $y-$axis and the line $x+y=6$.

4. (10 marks) The temperature at each point of the sphere $x^2 + y^2 + z^2 = 11$ is given by the function $T(x, y, z) = 20 + 2x + 2y + z^2$. Find the highest and lowest temperature on the intersection of the given sphere $x^2 + y^2 + z^2 = 11$ and the plane $x + y + z = 3$.

5. (10 marks) Using ONE iteration of Newton-Raphson's method with initial guess $(1, -1)$ to solve the system of linear equations

$$\begin{cases} x^2 + 4y^2 - 9 = 0 \\ 18y - 14x^2 + 45 = 0 \end{cases}.$$

6. (10 marks) Evaluate $\iint_R (x^2 - xy - 2y^2)\, dx\, dy$, where R is the region in the xy-plane bounded by the lines $x - 2y = 0$, $x - 2y = 1$, $x + y = 0$ and $x + y = 1$.

7. (10 marks) Find the volume of the solid region A between $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 = ax$ $(a > 0)$.

8. (10 marks) Find the volume of the solid region B in the first octant between $x + y + z^2 = 1$, $x = 0$, $y = 0$ and $z = 0$.

9. (10 marks) Given that

$$A = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix P and a diagonal matrix D, such that $A = PDP^{-1}$.

(c) Find A^{2018}.

10. (10 marks) Read the following information and answer the questions.

Many types of applications involve a finite set of states $\{S_1, S_2, \cdots S_n\}$ of a given population. For instance, residents of Hong Kong may live in Hong Kong Island, Kowloon or New Territories. The probability that a member of a population will change from the $j-$th state to the $i-$th state is represent by a number p_{ij}, where $0 < p_{ij} < 1$. A probability of $p_{ij} = 0$ means that the member is certain not to change from the $j-$th state to the $i-$th state. P is called the matrix of transition probabilities because it gives the probabilities of each possible type of transition (or change) within the population.

$$P = \begin{array}{c} \overbrace{\begin{array}{cccc} S_1 & S_2 & \cdots & S_n \end{array}}^{From} \\ \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} \end{array} \begin{array}{c} \\ S_1 \\ S_2 \\ \vdots \\ S_n \end{array} \quad To$$

At each transition, each member in a given state must either stay in that state or change to another state. For probabilities, this means that the sum of the entries in any column of P is 1. In general, such a matrix is called "stochastic". That is, an $n \times n$ matrix P is called a stochastic matrix if each entry is a number bewteen 0 and 1 and each column of P adds up to 1.

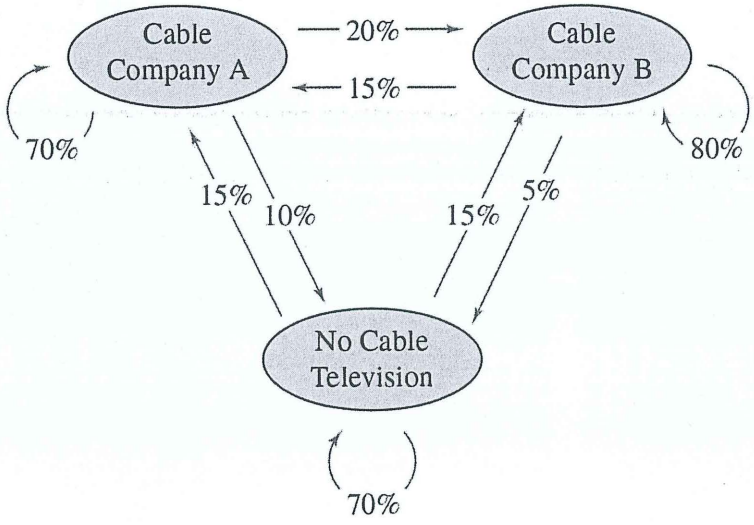
(a) Determine whether the following matrices are stochastic matrices.

$$A = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.2 & 0.3 & 0.4 \\ 0.3 & 0.4 & 0.5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1/2 & 1/3 & 1/4 \\ 1/4 & 0 & 3/4 \\ 1/4 & 2/3 & 0 \end{bmatrix}$$

(b) Two competing companies in Hong Kong offer cable television service to 100,000 households. The changes in cable subscriptions each year are

shown in the following figure. Company A now has 15,000 subscribers and Company B has 20,000 subscribers. Write the matrix P of transition probabilities and determine how many subscribers will each company have 1 year from now?



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